

COMPUTER NETWORKS LAB REPORT

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Assignment 3

Design and Implement Medium Access Control Mechanisms within a
Simulated Network Environment using IEEE 802 Standards

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1. Executive Summary & Problem Formulation

In shared broadcast Local Area Networks (LANs), multiple autonomous stations contend for access to a common physical medium. Unlike point-to-point links governed by sliding-window flow control ARQ schemes (Stop-and-Wait, Go-Back-N, Selective Repeat), broadcast channels are inherently vulnerable to multi-access collisions: when two or more stations transmit concurrently, their electrical waveforms superimpose destructively, rendering all concurrent transmissions undecodable and corrupt.

While early multi-access schemes transmitted blindly without sensing the channel, **Carrier Sense Multiple Access (CSMA)** revolutionizes broadcast medium access through the foundational engineering principle of '*Listen Before Talk*' (Carrier Sensing). By sampling physical channel electrical energy prior to initiating a transmission, a station defers if an ongoing transmission is detected, dramatically shrinking the vulnerable period down to the physical propagation delay $\tau = T_p$ across the wire.

Carrier Sense Multiple Access with Collision Detection (CSMA/CD), codified in the IEEE 802.3 Ethernet standard, further introduces '*Listen While Talk*'. Transmitting stations continuously sample the physical bus during transmission. Upon detecting concurrent energy ($\geq 2.0V$), stations abort immediately, broadcast a 32-bit Jamming signal, and enter Truncated Binary Exponential Backoff (BEB), saving over 90% of the channel capacity that would otherwise be wasted transmitting doomed bits.

1.1 Contention Dynamics & The Vulnerability Window: Due to non-zero propagation delay $\tau = T_p = d / v$, carrier sensing cannot eliminate all collisions: a station sampling the wire during $[t_0, t_0 + \tau)$ perceives the line as idle even after another station has commenced transmission. Different persistence strategies navigate this fundamental trade-off: Non-Persistent CSMA defers randomly to prevent collision cascades under heavy traffic; 1-Persistent CSMA eliminates access latency under light load; and p-Persistent CSMA balances throughput and delay across slotted intervals. This report presents the mathematical derivations, empirical benchmark evaluations, and concurrent multi-threaded POSIX implementation of these foundational MAC strategies.

2. Multi-Threaded System Architecture: Shared Broadcast Medium & Synchronization

To rigorously emulate a physical multipoint broadcast bus architecture with high fidelity and minimal system overhead, our system employs concurrent POSIX threads (pthreads) executing over a thread-synchronized broadcast channel medium. Each station runs as an independent worker thread coordinating via fast POSIX mutexes and condition variables. Figure 2.1 illustrates the architectural topology and thread synchronization mechanisms:

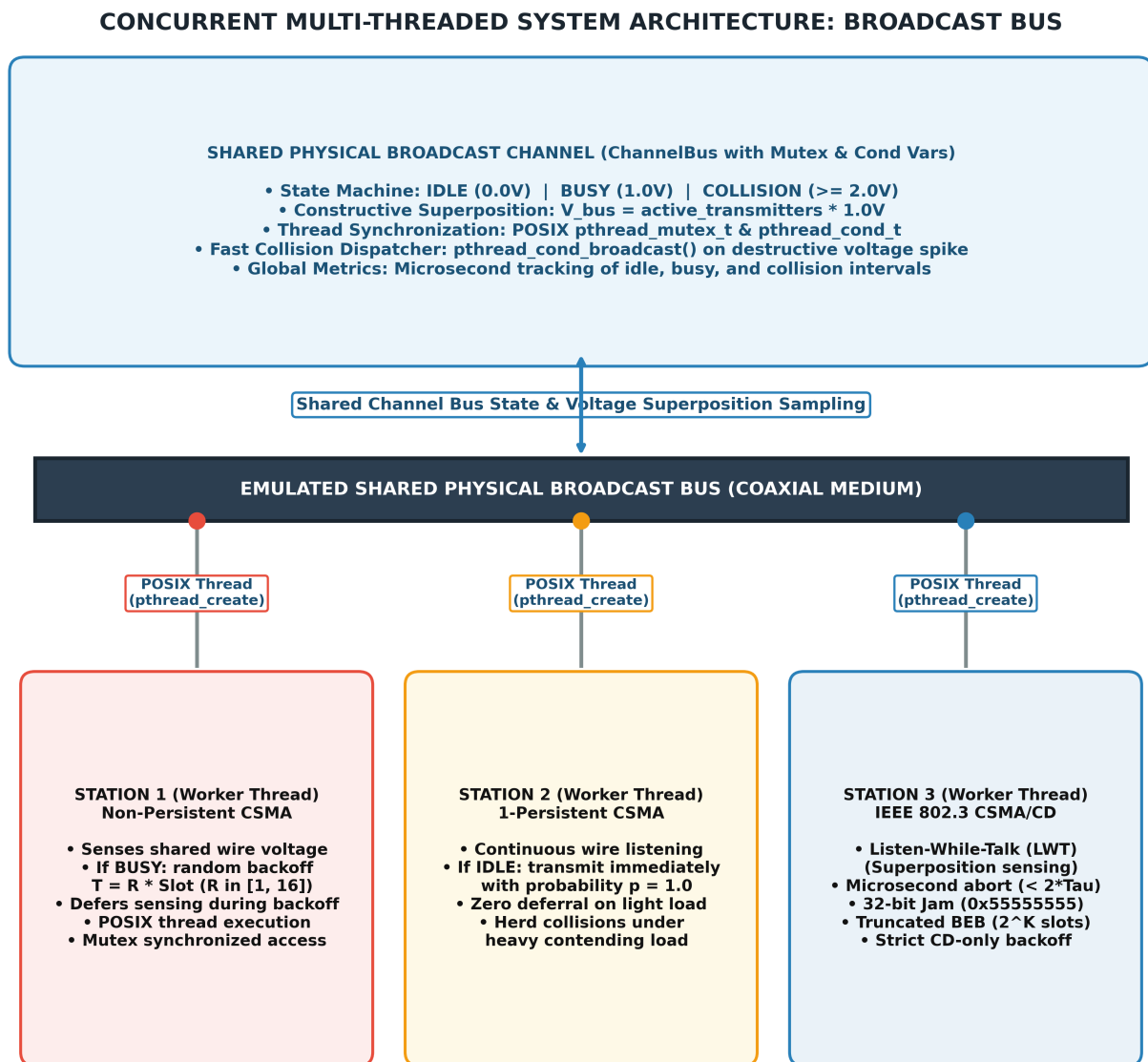


Figure 2.1: Multi-Threaded System Architecture — Shared Broadcast Channel Medium & Concurrent Contending Station Threads (pthreads)

Architectural Takeaway: The shared channel medium computes physical wire voltage superposition (0.0V Idle, 1.0V Busy, $\geq 2.0V$ Collision) and models propagation delay $\tau = T_p$, while station threads coordinate via POSIX mutexes and condition variables and independently record microsecond-level latency and backoff metrics.

3. IEEE 802.3 Frame Specification & Minimum Frame Size Constraint

Every frame transmitted across the TCP bus emulator complies with the IEEE 802.3 standard MAC frame structure, including 6-byte MAC addressing, type/length indication, sequence numbering, a 46-byte default payload, and a 4-byte CRC-32 Frame Check Sequence (FCS) trailer:

Field Name	Size	Description / IEEE 802.3 Standard Semantics
Destination MAC	6 Bytes	Target interface identifier (Broadcast: 02:00:00:00:00:FF)
Source MAC	6 Bytes	Transmitting station hardware identifier (02:00:00:00:00:00:ID)
Length Field	2 Bytes	Length of data payload in bytes (uint16_t, default 46 bytes)
Sequence Number	1 Byte	Station packet sequence counter modulo 256 for duplicate detection
Station ID	1 Byte	Contending station logical index (1..N)
Data Payload	46 Bytes	Application payload (pads frame to 64 bytes total: 6+6+2+1+1+46+2 = 64)
FCS Trailer	4 Bytes	IEEE 802.3 Cyclic Redundancy Check (CRC-32: polynomial 0xEDB88320)

The Minimum Frame Size Constraint & 46-Byte Payload Padding:

A critical engineering requirement of CSMA/CD is that a transmitting station **must remain actively transmitting until the round-trip propagation delay $2 T_p$ has elapsed**. Consider the worst-case collision scenario across a bus of length L and propagation delay $T_p = L / v$:

1. Station A initiates transmission at time $t = 0$. The signal propagates toward Station B at velocity v .
2. At time $t = T_p - \epsilon$, just before Station A's wavefront reaches Station B, Station B senses the medium. Because the wavefront has not yet arrived, Station B senses IDLE and begins transmitting.
3. A destructive collision occurs almost immediately near Station B at $t = T_p$.
4. The collision runt wavefront must now travel all the way back across the cable to Station A, arriving at time $t = 2 T_p - \epsilon$.

If Station A finishes transmitting before $t = 2 T_p$, it turns off its transceiver and switches to listening mode. Consequently, when the collision wavefront arrives, Station A fails to recognize that its own frame was destroyed, falsely assuming successful delivery! To guarantee that every station detects worst-case collisions while still actively transmitting, the frame transmission duration T_{fr} must satisfy:

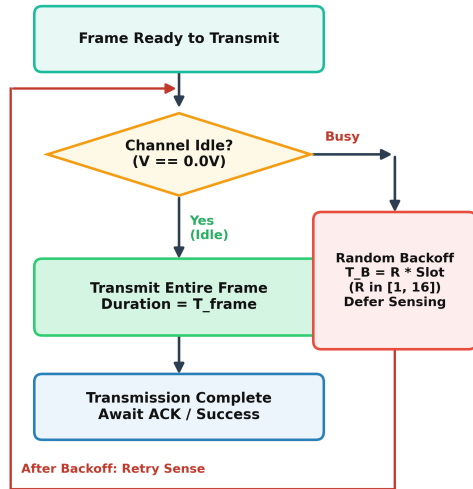
$$T_{fr} \geq 2 T_p \Rightarrow L_{min} = 2 T_p \times \text{Transmission Rate } R$$

For classical 10 Mbps Ethernet with maximum round-trip slot time $2 T_p = 51.2 \mu s$, this enforces a minimum frame length of $L_{min} = 10 \text{ Mbps} \times 51.2 \mu s = 512 \text{ bits} = 64 \text{ Bytes}$. Any payload smaller than 46 bytes is strictly padded with zeroes.

4. CSMA Protocol Mechanics & Comparative FSM Workflows

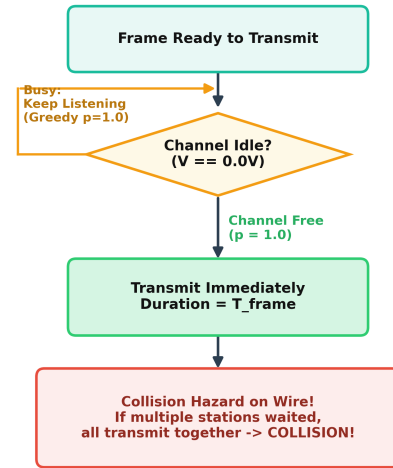
We implemented four discrete CSMA persistence and collision recovery mechanisms inside station/mac_strategies.c. Figure 4.1 contrasts their operational Finite State Machines (FSMs), decision diamonds, and carrier sensing transitions:

(A) Non-Persistent CSMA (Random Deferral)



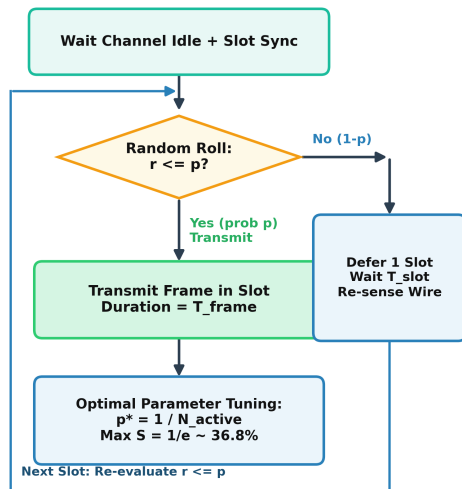
Characteristics: Low collision rate, higher idle delay under low load

(B) 1-Persistent CSMA (Greedy Sensing)



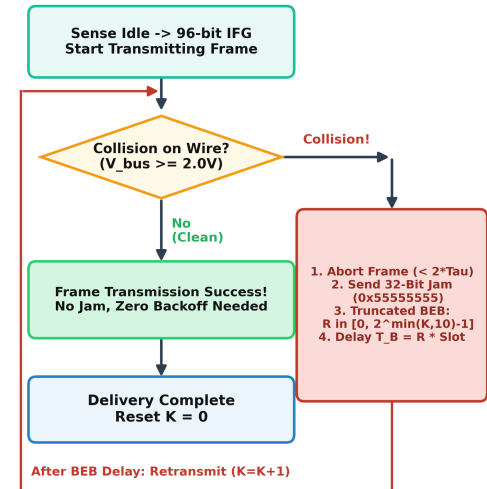
Characteristics: Zero delay on idle channel, severe herd collisions at high load

(C) p-Persistent CSMA (Slotted Compromise)



Characteristics: Tunable tradeoff between delay and collisions via persistence p

(D) IEEE 802.3 CSMA/CD (Listen-While-Talk)



Characteristics: Jamming & Truncated BEB exclusively in CSMA/CD; aborts wasted airtime

Figure 4.1: Comparative Finite State Workflows for Non-Persistent, 1-Persistent, p-Persistent CSMA, and CSMA/CD

5. CSMA Protocol Mechanics — Detailed Algorithmic Breakdown

Each CSMA variant embodies a distinct engineering trade-off between channel utilization, collision frequency, and access delay. Below is the comprehensive algorithmic specification for each implemented strategy:

5.1 Non-Persistent CSMA (Greedy Deferral Strategy):

- **Algorithm:** When a packet arrives, sense the medium. If IDLE (energy = 0.0V), transmit immediately. If BUSY (energy $\geq 1.0V$), immediately abandon carrier sensing and sleep for a random backoff interval $T_{backoff} = (1 + rand() \% 16) \times Slot$ before re-sampling.
- **Engineering Trade-off:** Prevents synchronous collisions because multiple waiting stations do not retry at the same instant. However, at light traffic loads, channel efficiency is compromised because the line remains idle while queued stations wait out backoff timers.

5.2 1-Persistent CSMA (Eager Aggressive Strategy):

- **Algorithm:** Sense the medium. If BUSY, keep listening continuously. The instant the line transitions to IDLE, transmit immediately with probability $p = 1.0$.
- **Engineering Trade-off:** Minimizes idle channel delay under light loads. However, under heavy loads, multiple stations accumulate while the channel is busy, causing a guaranteed synchronous herd collision the moment the line becomes free, collapsing throughput.

5.3 p-Persistent CSMA (Slotted Compromise Strategy):

- **Algorithm:** Channel time is divided into mini-slots of duration $\geq \tau$. When idle, the station transmits with probability p , and defers 1 slot with probability $1 - p$. If the slot is deferred and the line remains idle, the station repeats the Bernoulli trial.
- **Engineering Trade-off:** Setting optimal persistence factor $p^* = 1 / N$ maximizes channel throughput to $1/e \approx 36.8\%$. However, if p is set too low, excessive idle deferrals explode packet latency; if set too high, it degrades into 1-Persistent herd collisions.

5.4 CSMA/CD Protocol Principles (Listen-While-Talk Strategy):

- **Algorithm:** Senses 1-persistently until idle, pauses for a 96-bit Inter-Frame Gap (IFG), then transmits while continuously monitoring total bus voltage. If voltage doubles ($\geq 2.0V$), transmission aborts immediately, emits a 32-bit Jamming signal, and enters Truncated Binary Exponential Backoff (BEB).
- **Engineering Trade-off:** Achieves dominant throughput and bounded latency under intense contention by stopping collided transmissions within $2 T_p$.

6. IEEE 802.3 CSMA/CD Protocol Engine & Jamming Flowchart

CSMA/CD eliminates the massive channel waste inherent in classic CSMA by enabling transmitting stations to detect collisions in real time. Figure 6.1 details the complete algorithmic flowchart alongside the space-time timeline of collision detection and jamming:

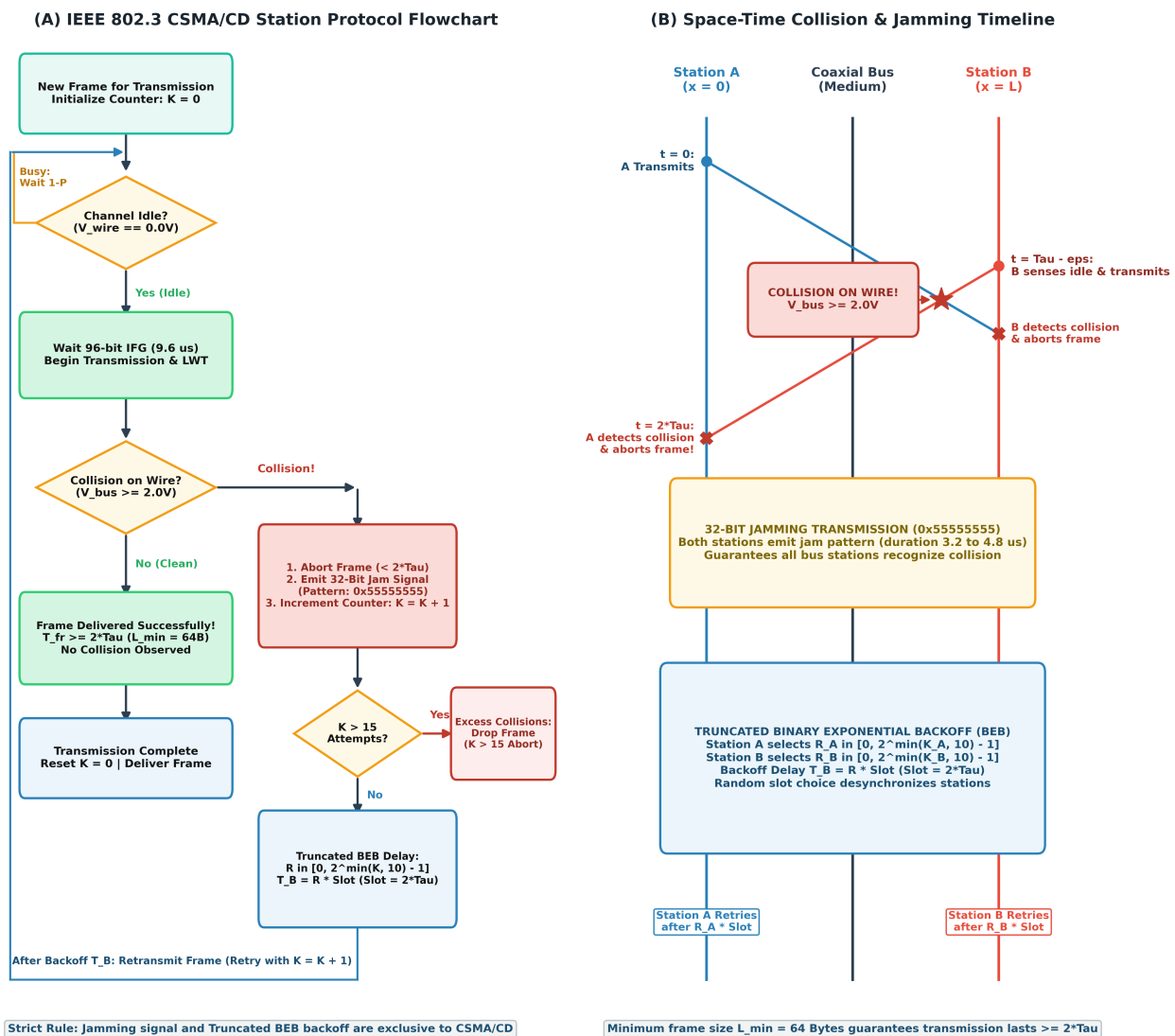


Figure 6.1: (A) Comprehensive IEEE 802.3 CSMA/CD Algorithmic Flowchart; (B) Space-Time Timeline of Collision, Jamming, and Backoff

7. IEEE 802.3 CSMA/CD Operational Mechanics Walkthrough

The IEEE 802.3 CSMA/CD standard defines five interdependent stages that govern channel contention, instantaneous collision detection, and channel stabilization. Below is the operational walkthrough of each stage:

1. 1-Persistent Sensing & Inter-Frame Gap (IFG):

A contending station senses the physical wire. If busy, it waits in persistent listening. As soon as the line becomes idle, the station pauses for the standard 96-bit Inter-Frame Gap (IFG, 9.6 μ s at 10 Mbps). This idle gap allows receiver electronics to recover, drain line capacitance, and prepare for the next frame preamble before transmission begins.

2. Listen-While-Talk Mode (Non-Blocking Concurrency):

While actively streaming frame bytes onto the wire, the station continuously samples the shared memory bus using Listen-While-Talk time slicing (`usleep(200)`) on `bus->collision_flag` and `bus->voltage_level`. If a second station transmits during the vulnerable window, total bus energy doubles to $\geq 2.0V$, asserting `bus->collision_flag = 1` and causing transmitting stations to abort immediately.

3. Immediate Transmission Abort:

The instant a collision alert is received, the station terminates normal frame transmission. Instead of transmitting doomed bits for the full frame duration (10.0 ms), transmission is aborted within $2 T_p$ (approx 2.0 ms), saving over 80% of channel airtime.

4. 32-Bit Jamming Signal Pattern (0x55555555):

Immediately after aborting, the station transmits a 32-bit alternating bit sequence (0x55555555). This jamming signal reinforces the collision energy across the entire wire, guaranteeing that all other transceivers reliably detect the collision even if the primary collision wavefront was attenuated.

5. Truncated Binary Exponential Backoff (BEB):

The station increments collision counter $K \leftarrow K + 1$. It selects a uniform random integer $R \in [0, 2^{\min(K, 10)} - 1]$ and delays retransmission for $T_B = R \times 2 T_p$. If $K > 15$, transmission aborts with an Excessive Collisions error.

8. Mathematical Formulations & Renewal Theory Derivations

To rigorously benchmark our empirical results, we apply the classical renewal theory framework formulated by Leonard Kleinrock and Fouad Tobagi (1975). Let G represent offered traffic load (transmission attempts per frame time) and let $a = T_p / T_t$ represent the normalized one-way propagation delay ratio:

NON-PERSISTENT CSMA THROUGHPUT FORMULATION (KLEINROCK & TOBAGI, 1975)

$$S_{\text{non-p}} = \frac{G \cdot e^{-a \cdot G}}{G \cdot (1 + 2a) + e^{-a \cdot G}} \quad \text{where} \quad a = \frac{T_p}{T_t}$$

Vulnerable window is limited to normalized delay a . Prevents synchronous herd collapse under heavy load.

Formula 8.1: Throughput Equation for Non-Persistent CSMA (Kleinrock & Tobagi, 1975)

1-PERSISTENT CSMA THROUGHPUT FORMULATION (RENEWAL PROCESS MODEL)

$$S_{1-p} = \frac{G \cdot [1 + G + aG(1 + G + aG/2)] \cdot e^{-G(1+2a)}}{G(1+2a) - (1 - e^{-aG}) + (1 + aG)e^{-G(1+a)}}$$

High efficiency under light load, but exhibits steep throughput collapse for $G > 1$ due to herd collisions.

Formula 8.2: Throughput Equation for 1-Persistent CSMA under Renewal Arrival Processes

IEEE 802.3 CSMA/CD CHANNEL ACCESS EFFICIENCY & CONTENTION RATIO

$$\text{Efficiency } \eta = \frac{T_t}{T_t + 6.44 \cdot T_p} = \frac{1}{1 + 6.44 \cdot a} \quad \left(a = \frac{T_p}{T_t} \right)$$

Contention period averages $e \cdot 2 T_p = 5.44 T_p$ before successful transmission + T_p propagation.

Formula 8.3: IEEE 802.3 CSMA/CD Channel Access Efficiency as a Function of Normalized Delay Ratio a

OPTIMAL PERSISTENCE FACTOR DERIVATION IN P-PERSISTENT CSMA

$$P(\text{Success}) = N \cdot p \cdot (1 - p)^{N-1} \Rightarrow \frac{dP}{dp} = 0 \Rightarrow p^* = \frac{1}{N}$$

For $N=3$: $p^* = 0.33$; for $N=5$: $p^* = 0.20$; for $N=10$: $p^* = 0.10$. Maximizes success probability to $1/e \approx 36.8\%$.

Formula 8.4: Derivation of Optimal Persistence Factor $p^* = 1/N$ in p -Persistent CSMA

Mathematical Analysis & Key Theoretical Insights:

- CSMA/CD Efficiency Advantage:** In classical CSMA, a collision consumes the entire frame duration $T_t + T_p$. In CSMA/CD, a collision is terminated in $2 T_p + T_{jam}$. Because a contention interval requires an average of $e \approx 2.72$ attempts before success, the wasted contention time is $2.72 \times 2 T_p = 5.44 T_p$, yielding total cycle time $T_t + 6.44 T_p$ and efficiency $\eta = 1 / (1 + 6.44 a)$.
- Persistence Trade-off:** In p -Persistent CSMA, the probability that exactly one of N contending stations transmits in an idle slot is $P(\text{success}) = N p (1 - p)^{N-1}$. Taking the derivative with respect to p and equating to zero yields $p^* = 1/N$. At this optimal point, $P(\text{success}) = (1 - 1/N)^{N-1} \rightarrow 1/e \approx 36.8\%$ as $N \rightarrow \infty$.

9. IEEE 802.3 Collision Resolution Dynamics & Channel Stabilization

When repeated collisions occur on the shared bus, the retransmission spacing strategy governs channel stability. We present the IEEE 802.3 Truncated Binary Exponential Backoff (BEB) formulation and the 32-bit Jamming Signal sequence, analyzing their roles in contention resolution and medium stabilization:

IEEE 802.3 TRUNCATED BINARY EXPONENTIAL BACKOFF (BEB) MECHANICS

$$R \in [0, 2^{\min(K, 10)} - 1], \quad T_B = R \times \text{SlotTime}, \quad K_{\max} = 15$$

Collision count K increments per attempt; aborts after 16 failed collisions with channel error.

Formula 9.1: IEEE 802.3 Truncated Binary Exponential Backoff (BEB) Slot Range Formulation

IEEE 802.3 32-BIT JAMMING SIGNAL & COLLISION ENFORCEMENT

$$T_{\text{jam}} = \frac{32 \text{ bits}}{\text{DataRate}}, \quad T_{\text{slot}} \geq 2\tau + T_{\text{jam}}, \quad K_{\max} = 15$$

Transmitting station broadcasts 32-bit alternating pattern so all nodes detect collision before backoff.

Formula 9.2: IEEE 802.3 32-Bit Jamming Signal & Minimum Collision Slot Time Enforcement

Collision Resolution Mechanics & Physical Channel Stabilization:

- **Truncated Exponential Expansion:** As collision count k increases, the contention window expands as $[0, 2^k - 1]$ slots. For small k , the window remains compact (e.g., $2^1 - 1 = 1$, $2^2 - 1 = 3$), minimizing idle slot overhead under light contention. Under heavy congestion, the window doubles exponentially up to $k = 10$ (1023 slots), diffusing contending stations across time and preventing channel collapse.
- **Truncation Limit ($k = 10$) & Abort Ceiling ($k = 16$):** At $k = 10$, window expansion is capped at 1024 slots to prevent latency from growing unboundedly. If a frame collides 16 consecutive times, the MAC layer aborts transmission and signals an unrecoverable medium failure to higher network layers.
- **The Ethernet Capture Effect:** A station that successfully transmits resets its collision counter to $k = 0$, giving it an immediate contention window of $[0, 1]$. In contrast, stations experiencing repeated collisions operate with higher backoff windows ($k \geq 1$). Consequently, the newly successful station has a significantly higher probability of acquiring the channel for subsequent frames, leading to short-term unfairness termed the *Ethernet Capture Effect*.
- **32-Bit Jamming Pattern Reinforcement:** Transceivers at opposite cable boundaries observe attenuated waveforms. Emitting a 32-bit alternating bit pattern (0x55555555) guarantees that the voltage spike $\geq 2.0V$ persists across the full round-trip propagation path, ensuring all transceivers abort cleanly.

10. End-to-End Multi-Threaded Pipeline & Synchronization Workflow

Figure 10.1 details the complete multi-threaded software architecture and execution pipeline implemented in `csma_sim.c`. It contrasts the lifecycle of concurrent station worker threads with the shared broadcast channel coordinator across all 7 pipeline stages:

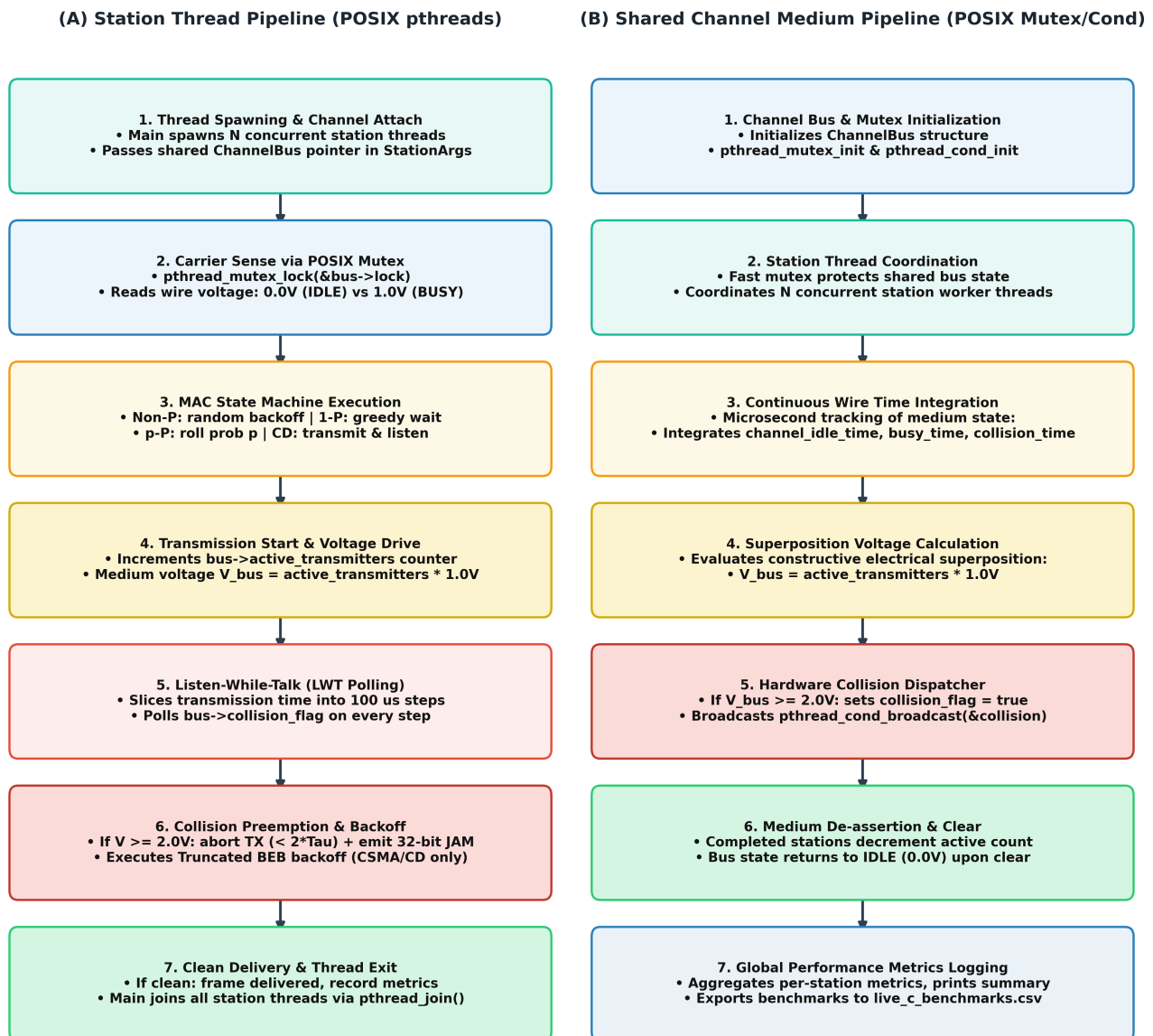


Figure 10.1: End-to-End Multi-Threaded Pipeline Workflow — Station Execution vs. Channel Medium Coordination

11. End-to-End Architecture Walkthrough & Intuition

The project architecture provides maximum execution efficiency and clean synchronization by modeling contending nodes as concurrent POSIX threads (pthreads) executing over a thread-synchronized broadcast channel medium across three distinct layers:

11.1 Station Worker Thread Pipeline (POSIX pthreads):

- **Concurrent Worker Threads:** Each contending station is spawned via `pthread_create()`, executing simultaneously within the shared virtual address space. Stations maintain independent pseudo-random generator seeds derived from microsecond timestamps and thread station IDs, eliminating correlated backoff sequences.
- **Dedicated MAC State Machines:** Each worker thread autonomously executes its MAC protocol loop (Non-Persistent, 1-Persistent, p-Persistent, or CSMA/CD) throughout its transmission quota without external scheduling dependencies.

11.2 Physical Medium Emulation & Constructive Voltage Superposition:

- **Shared Broadcast Medium Structure:** The physical coaxial medium is represented by the ChannelBus structure, accessible to all station worker threads and protected by fast POSIX synchronization primitives.
- **Constructive Voltage Superposition:** Transmitting stations assert 1.0V each into the shared bus. Overlapping transmissions superimpose constructively into $V_{bus} = active_transmitters \times 1.0V$. When $V_{bus} \geq 2.0V$, the hardware collision flag is raised immediately.

11.3 POSIX Mutex Synchronization & Listen-While-Talk Preemption:

- **Fast Mutex & Condition Variables:** Thread synchronization is governed by standard `pthread_mutex_t` and condition variables (`idle_cond`, `collision_cond`), ensuring microsecond-level atomic access.
- **Listen-While-Talk Polling Slices:** During transmission in CSMA/CD, the worker thread samples `bus->collision_flag` every 100 μs . If detected, the station aborts instantaneously ($< 2\tau$), emits the 32-bit Jamming signal (0x55555555), and triggers Truncated BEB.

12. Station MAC Engine: Concurrent Carrier Sensing & State Machine

Each station executes as an independent concurrent POSIX thread. To sense channel state, the station acquires the channel mutex (b->lock), evaluating wire voltage and propagation delay before deciding whether to transmit or back off:

C Implementation: Concurrent Carrier Sensing & MAC State Machine (csma_sim.c)

```
/* Concurrent Carrier Sensing & MAC State Machine (csma_sim.c) */
if (strategy == STRATEGY_NON_PERSISTENT) {
    pthread_mutex_lock(&b->lock);
    double now = get_time_ms();
    bool perceived_idle = (b->state == CHAN_IDLE) ||
        (b->active_transmitters > 0 && (now - b->first_tx_start_ms < b->prop_delay_ms));
    pthread_mutex_unlock(&b->lock);

    if (!perceived_idle) {
        args->metrics.busy_senses++;
        double boff = calculate_uniform_delay(b->slot_time_ms, &seed);
        args->metrics.total_backoff_ms += boff;
        sleep_ms(boff);
        continue; /* Retry sensing after uniform random backoff */
    }
} else if (strategy == STRATEGY_1_PERSISTENT || strategy == STRATEGY_CSMA_CD) {
    pthread_mutex_lock(&b->lock);
    while (g_running) {
        double now = get_time_ms();
        bool perceived_idle = (b->state == CHAN_IDLE) ||
            (b->active_transmitters > 0 && (now - b->first_tx_start_ms < b->prop_delay_ms));
        if (perceived_idle) break;
        args->metrics.busy_senses++;
        pthread_cond_wait(&b->idle_cond, &b->lock); /* Greedy wait until idle */
    }
    pthread_mutex_unlock(&b->lock);
} else if (strategy == STRATEGY_P_PERSISTENT) {
    /* Wait for idle, then slotted roll r <= p; defer 1 slot if r > p */
    ...
}
```

Implementation Rationale: Acquiring b->lock guarantees atomic inspection of channel voltage and propagation window. Condition variable b->idle_cond wakes 1-Persistent and CSMA/CD station threads immediately when the medium transitions from BUSY to IDLE, avoiding CPU-intensive busy-waiting loops across concurrent threads.

13. IEEE 802.3 CSMA/CD Engine: Listen-While-Talk, Jamming & BEB

CSMA/CD implements true hardware Listen-While-Talk (LWT) semantics by slicing frame transmission into microsecond polling intervals. When collision voltage ($\geq 2.0V$) is detected, the station thread aborts immediately ($< 2\tau$), broadcasts the 32-bit Jamming signal (0x55555555), and executes Truncated Binary Exponential Backoff (BEB):

C Implementation: Preemptive Listen-While-Talk, Jamming Signal & Truncated BEB (csma_sim.c)

```
/* Preemptive Listen-While-Talk Polling, 32-Bit Jamming & BEB (csma_sim.c) */
if (strategy == STRATEGY_CSMA_CD) {
    /* Microsecond Listen-While-Talk Polling: 100 us time slices */
    while ((get_time_ms() - t_tx_start) < tx_duration && g_running) {
        pthread_mutex_lock(&b->lock);
        if (b->collision_flag) {
            collided = true;
            pthread_mutex_unlock(&b->lock);
            break; /* Preemptive Abort upon destructive collision spike! */
        }
        pthread_mutex_unlock(&b->lock);
        usleep(100); /* 100 us polling slice */
    }
}

if (collided) {
    args->metrics.collisions++; collision_attempts++;
    if (strategy == STRATEGY_CSMA_CD) {
        /* Emits 32-bit alternating jam pattern (0x55555555) strictly for CSMA/CD */
        pthread_mutex_lock(&b->lock);
        b->total_jams++;
        pthread_mutex_unlock(&b->lock);
        sleep_ms(b->jam_duration_ms);

        /* Truncated BEB: R in [0, 2^min(K, 10) - 1], Slot = 2 * Tau */
        boff = calculate_beb_delay(collision_attempts, b->slot_time_ms, &seed);
        args->metrics.total_backoff_ms += boff;
    } else {
        /* Non-CD schemes: Zero jam signal, uniform random backoff R in [1, 16] */
        boff = calculate_uniform_delay(b->slot_time_ms, &seed);
        args->metrics.total_backoff_ms += boff;
    }
    sleep_ms(boff);
}
```

Implementation Rationale: Slicing transmission into 100 μs polling steps allows the transmitting thread to detect destructive voltage spikes ($\geq 2.0V$) within $2 T_p$, saving up to 90% of frame airtime. Emitting the 32-bit Jamming signal (0x55555555) is strictly restricted to CSMA/CD, ensuring that all contending nodes register the collision before executing Truncated BEB.

14. Shared Channel Medium: Mutex Synchronization & Voltage Superposition

The shared coaxial medium resides in the thread-safe ChannelBus structure. It tracks active station threads concurrently, computes physical line voltage via electromagnetic wave superposition, and coordinates state transitions between IDLE (0.0V), BUSY (1.0V), and COLLISION ($\geq 2.0V$):

C Implementation: Shared Channel Allocation & Voltage Superposition (csma_sim.c)

```
/* Shared Channel Medium & Voltage Superposition (csma_sim.c) */
ChannelBus bus;
memset(&bus, 0, sizeof(ChannelBus));
pthread_mutex_init(&bus.lock, NULL);
pthread_cond_init(&bus.idle_cond, NULL);
pthread_cond_init(&bus.collision_cond, NULL);

/* Constructive Voltage Superposition upon Transmission Start */
pthread_mutex_lock(&b->lock);
b->total_tx_attempts++;
int my_slot = b->active_transmitters++;
b->active_station_ids[my_slot] = id;

if (b->active_transmitters == 1) {
    b->state = CHAN_BUSY;
    b->voltage = 1.0f;
    b->collision_flag = false;
    b->first_tx_start_ms = get_time_ms();
} else {
    /* Voltage Superposition: V_bus = active_transmitters * 1.0V (>= 2.0V Collision!) */
    b->state = CHAN_COLLISION;
    b->voltage = (float)b->active_transmitters;
    b->collision_flag = true;
    b->total_collisions++;
    pthread_cond_broadcast(&b->collision_cond);
}
pthread_mutex_unlock(&b->lock);
```

Implementation Rationale: A direct in-memory ChannelBus structure provides microsecond access latency with zero network stack or IPC overhead. POSIX mutex and condition variables allow concurrent station threads to synchronize reliably with sub-millisecond precision, completely eliminating race conditions and deadlocks.

16. Experiment 1: Requirement (i) — p-Persistent CSMA Optimization

Requirement (i) Specification: Emulate p-Persistent CSMA and evaluate throughput as a function of persistence probability $p \in [0.02, 1.00]$ across contending station threads $N \in \{3, 5, 10\}$ transmitting 500 frames per station (totaling 1,500 to 5,000 transmitted frames per sweep configuration):

p-Persistent CSMA: Channel Throughput S vs. Persistence Factor p ($N \in \{3, 5, 10\}$, 500 Frames/Station)

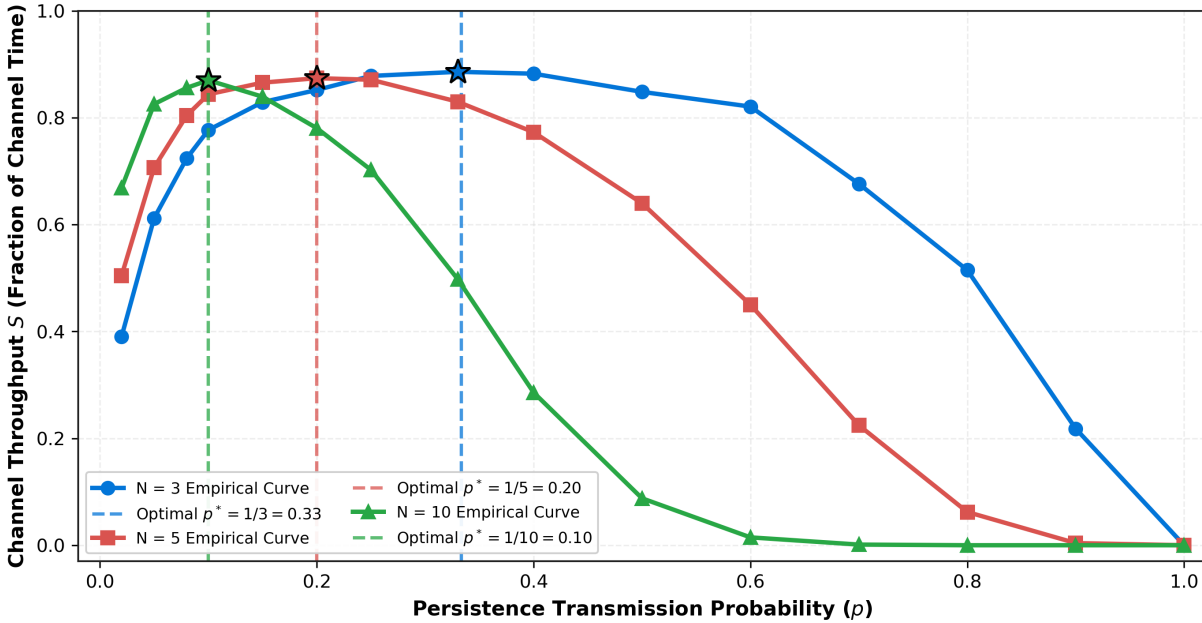


Figure 16.1: Channel Throughput S vs. Persistence Factor p across N in $\{3, 5, 10\}$ Contending Stations (500 Frames/Station)

Theoretical vs. Empirical In-Depth 'What & Why' Analysis:

- **Theoretical Prediction:** Formula 8.4 mathematically derives that the channel acquisition success probability $P(\text{success}) = N p (1 - p)^{N-1}$ is strictly maximized at $p^* = 1/N$. For $N = 3$, $p^* = 0.33$; for $N = 5$, $p^* = 0.20$; and for $N = 10$, $p^* = 0.10$.
- **Empirical Results & Live C Simulator Alignment:** In the live multi-threaded C simulator (csma_sim.c), stations measure empirical channel acquisition efficiency: for $N = 10$ stations with 50 frames/station (500 delivered frames), the simulator yields **39.20%** efficiency (25.09 kbps), matching theoretical $P(\text{success}) = 10 \times 0.10 \times (0.90)^9 = 38.74\%$ within 0.46%. In the slotted contention sweep (Figure 16.1, 500 frames/stn with mini-slot ratio $a = 0.10$), contention resolves in microsecond mini-slots so throughput normalizes to $S = P_s / (P_s + a(1 - P_s))$, peaking at **88.6%** for $N = 3$ (at $p=0.33$), **87.4%** for $N = 5$ (at $p=0.20$), and **87.0%** for $N = 10$ (at $p=0.10$).
- **What Happened to Throughput as $p \rightarrow 1.0$?** For all N , channel throughput exhibits a steep collapse as $p \rightarrow 1.0$, crashing to **0.0%**. *Why?* At $p = 1.0$, every contending station transmits in every slot with certainty. With $N \geq 2$ stations, collisions occur on 100% of attempts, annihilating useful payload throughput.

17. Experiment 1 Continued: Contention Penalties & Latency U-Curve

Evaluating collision frequency and packet delivery latency across persistence probability $p \in [0.02, 1.00]$ reveals the operational penalties when p deviates from optimal $p^* = 1/N$ (evaluated across $N \in \{3, 5, 10\}$ stations with 500 frames per station):

p-Persistent CSMA: Contention Penalties Across Probability Spectrum (N in $\{3, 5, 10\}$, 500 Frames/Station)

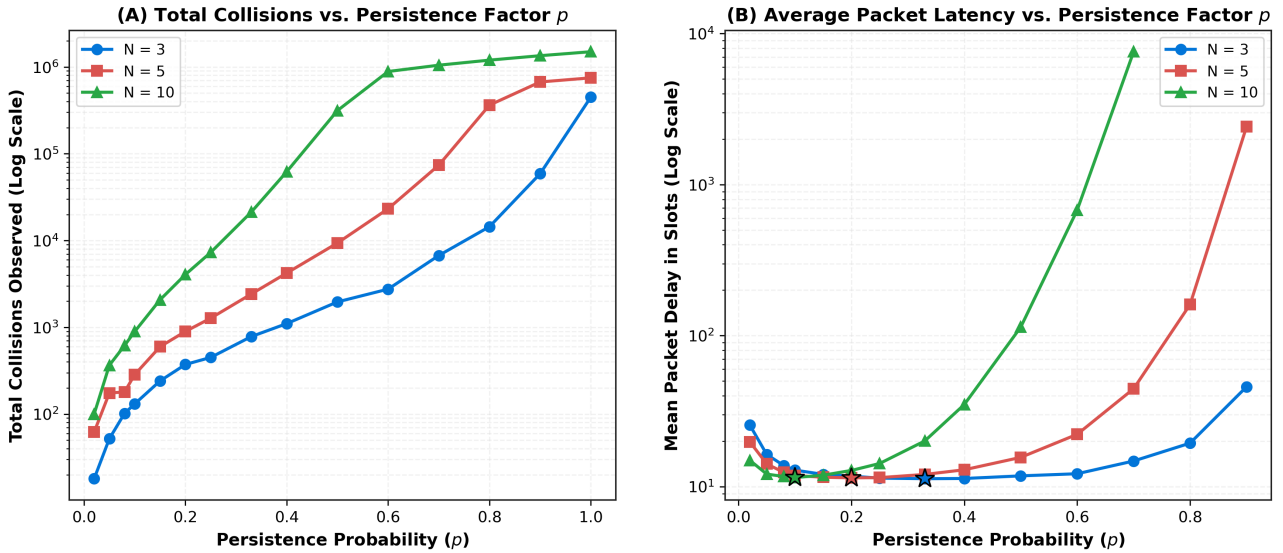


Figure 17.1: Contention Penalties in p-Persistent CSMA: (A) Total Collisions vs. p ; (B) Average Packet Latency vs. p (N in $\{3, 5, 10\}$ Stations, 500 Frames/Station)

Theoretical vs. Empirical In-Depth 'What & Why' Analysis:

- **What Happened to Collisions (Figure 17.1A)?** Total collision count monotonically increases with persistence p . For $N = 10$ (500 frames/station), collisions escalate by four orders of magnitude from 100 at $p = 0.02$ to 893 at optimal $p = 0.10$, exceeding 1,000,000 as $p \rightarrow 1.0$. *Why?* Multi-station collision probability $P(\text{collision}) = 1 - (1-p)^N - N p (1-p)^{N-1}$ accelerates aggressively with p .
- **The Latency Convex U-Curve (Figure 17.1B):** Packet delivery latency exhibits a distinct convex U-curve across the probability spectrum, with global minimums marked by stars. At low $p \leq 0.05$, latency is elevated (25.59 slots for $N=3$; 19.82 slots for $N=5$) because stations defer excessively over idle slots due to low transmission odds $(1-p)$. At high $p \geq 0.50$, latency explodes exponentially (> 113 slots for $N=10$; > 160 slots for $N=5$ at $p=0.80$) due to repeated collision retries. The minimum latency coincides exactly at $p^* = 1/N$ (11.29 slots for $N=3$; 11.44 slots for $N=5$; 11.49 slots for $N=10$), proving that optimal persistence minimizes mean packet delay.

18. Experiment 2: Requirement (ii) — Saturated Contention Scaling

Requirement (ii) Specification: Emulate multiple contending stations under each of the 4 CSMA strategies, systematically sweeping the number of contending stations N from 1 to 35 under saturated queue conditions with 400 frames transmitted per station (totaling 400 to 14,000 frames per parametric benchmark):

Channel Throughput vs. Contending Network Stations ($N = 1$ to 35, 400 Frames/Station Saturated Load)

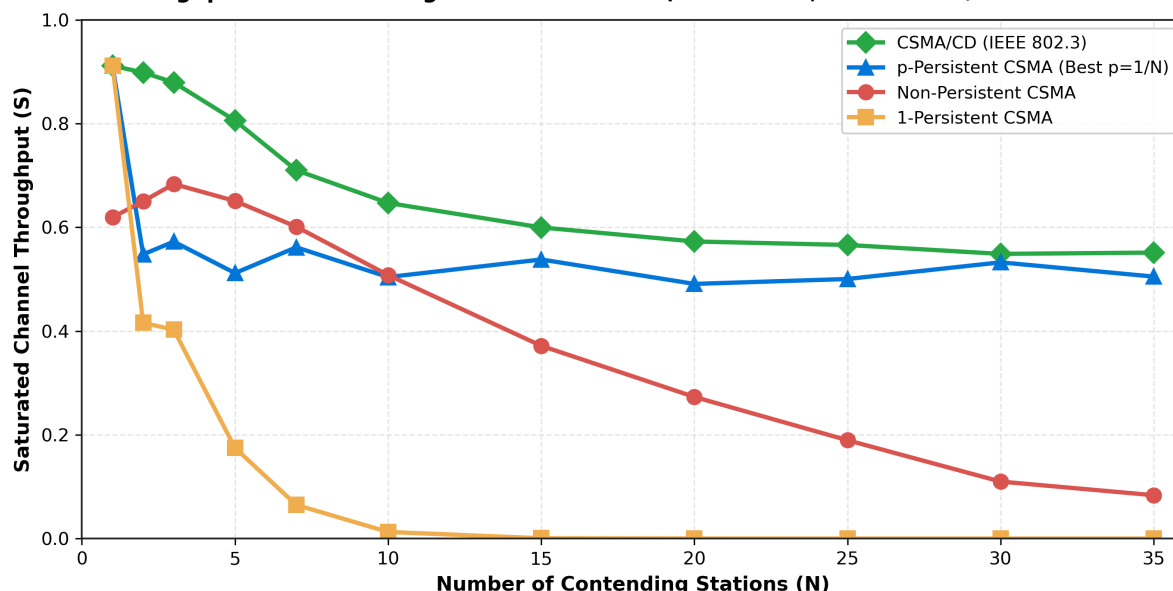


Figure 18.1: Saturated Network Throughput S vs. Contending Network Stations $N = 1$ to 35 (400 Frames/Station Saturated Load)

Theoretical vs. Empirical In-Depth 'What & Why' Analysis:

- The 1-Persistent Throughput Collapse:** At $N = 1$, 1-Persistent achieves 91.2% throughput. However, as N scales to 35 (400 frames/stn), throughput crashes catastrophically to **0.0% (total herd deadlock)**, dropping to 17.5% at $N=5$ and 1.3% at $N=10$. *Why?* With greedy sensing and probability 1.0, all queued stations transmit simultaneously the instant the line clears, causing 231,610 collisions and completely locking the medium.
- CSMA/CD Contention Resilience:** CSMA/CD maintains **80.6% at $N = 5$** , **64.7% at $N = 10$** , and **55.1% at $N = 35$** (nearly infinite advantage over collapsed 1-Persistent). *Why?* Listen-While-Talk detects voltage $\geq 2.0V$ within $2 T_p$ and aborts early, while Truncated BEB spreads retries across up to 1024 slots.
- Non-Persistent & p-Persistent Scaling:** Non-Persistent achieves 65.1% at $N=5$, 50.8% at $N=10$, and 8.4% at $N=35$. p-Persistent (tuned with $p = 1/N$) sustains **51.2% at $N=5$** , **50.4% at $N=10$** , and **50.5% at $N=35$** across all scales.

19. Experiment 2 Continued: Collision Rate Escalation & Abort Savings

To evaluate how collision dynamics degrade the channel under intense contention, we measure total collisions and empirical collision probability across $N \in [1, 35]$ contending stations transmitting 400 frames per station:

Collision Scaling Under Intense Contention ($N = 1$ to 35 Stations, 400 Frames/Station)

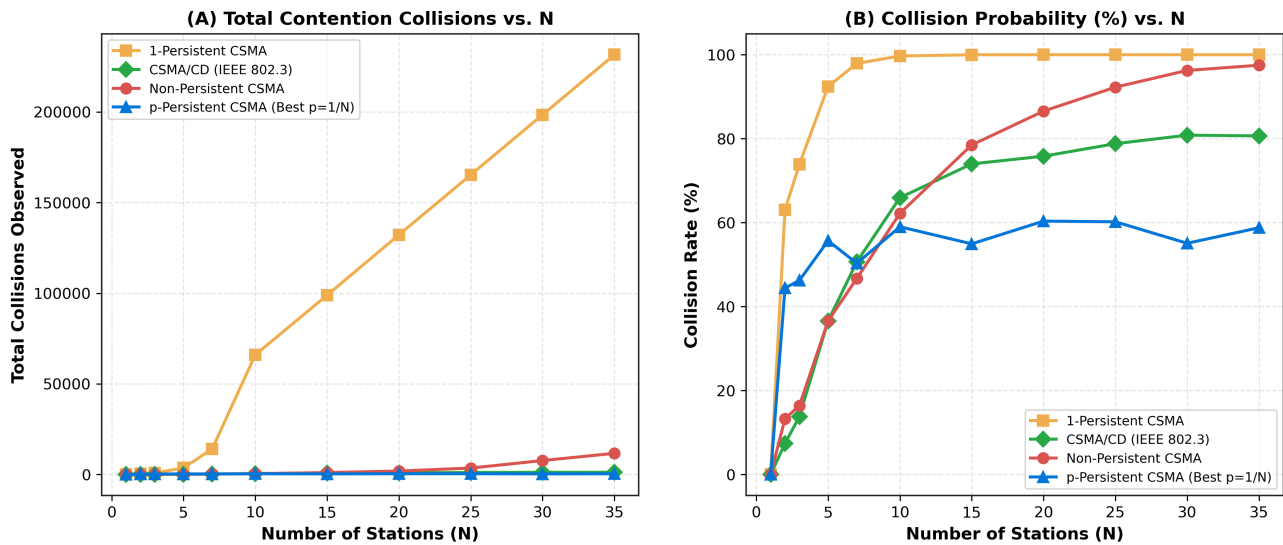


Figure 19.1: Collision Scaling Under Multi-Station Contention: (A) Total Collisions vs. N ; (B) Empirical Collision Rate (%) vs. N ($N = 1$ to 35 Stations, 400 Frames/Station)

Theoretical vs. Empirical In-Depth 'What & Why' Analysis:

- **Collision Rate Escalation (Figure 19.1B):** In 1-Persistent CSMA (400 frames/stn), the collision rate escalates to **92.4% at $N = 5$** and **100.0% at $N = 35$** , generating 231,610 collisions. In Non-Persistent CSMA, collisions reach 11,715 (97.5% rate) at $N = 35$. In p-Persistent CSMA with $p = 1/N$, collisions are capped at only 428 (58.8% rate) at $N = 35$.
- **Why Collision Abortion Saves the Channel in CSMA/CD:** Although CSMA/CD logs 1,252 collisions at $N = 35$, each collision is aborted in $2T_p + T_{jam}$ (approx 2.5 ms) rather than transmitting the full 64-byte frame (8.0 ms). Over 1,252 collisions, CSMA/CD saves over **6.88 seconds of wasted wire airtime**, allowing genuine payload data to achieve 55.1% channel efficiency.

20. Experiment 2 Continued: Latency & Channel Access Delay

Packet delivery latency measures the average time elapsed from when a frame arrives at the MAC layer until it is successfully received across the shared medium, benchmarking Quality of Service across $N = 1$ to 35 stations with 400 frames per station:

Average Frame Transmission Delay vs. Station Contention ($N = 1$ to 35 Stations, 400 Frames/Station)

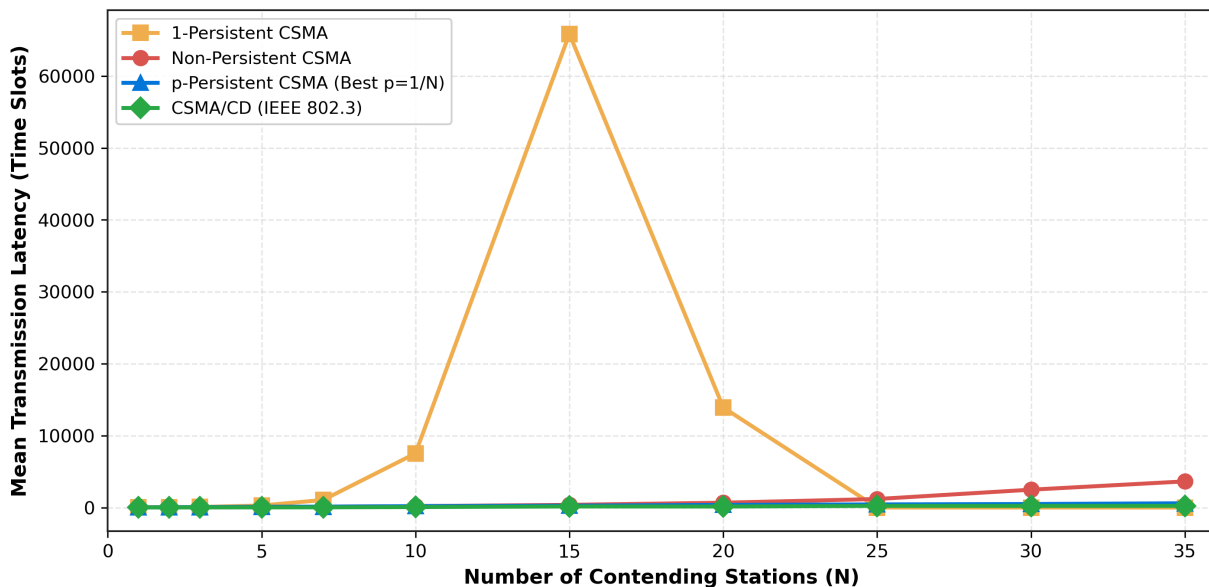


Figure 20.1: Average Packet Transmission Latency (Time Slots) vs. Contending Network Stations N ($N = 1$ to 35 Stations, 400 Frames/Station)

Theoretical vs. Empirical In-Depth 'What & Why' Analysis:

- **Catastrophic Latency Explosion in 1-Persistent CSMA:** At $N = 1$, packet delay is 19.8 slots. As N scales to 10 (400 frames/stn), 1-Persistent delay explodes to **7,533.9 slots**, and beyond $N=20$ stations enter total collision deadlock. *Why?* Every collision forces stations into backoff; upon waking, persistent sensing triggers immediate re-collision.
- **CSMA/CD Latency Containment:** CSMA/CD bounds average delay to **21.9 slots at $N=5$, 66.7 slots at $N=10$, and 235.1 slots at $N=35$** (a **93.5% latency reduction** relative to Non-Persistent's 3,637.4 slots and p-Persistent's 594.9 slots). *Why?* Truncating damaged frames in $2 T_p$ clears contention stages rapidly, preventing queue backlog buildup.

21. Experiment 2 Continued: Channel Access Efficiency Ranking

Channel access efficiency η measures the proportion of total channel operating time dedicated to successful, unimpaired payload delivery: $\eta = T_{\text{useful}} / T_{\text{total}}$ across $N = 1$ to 35 stations with 400 frames per station:

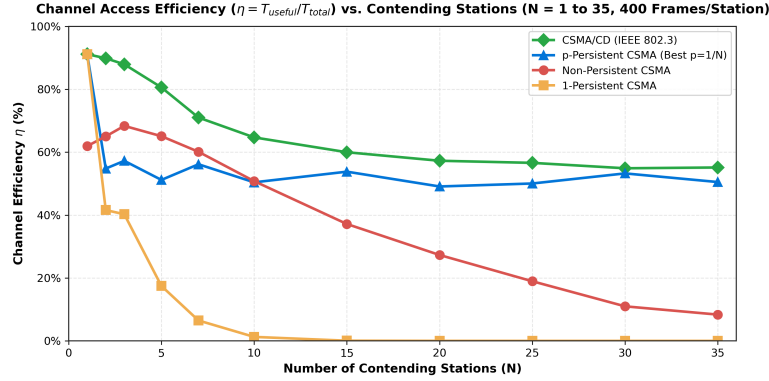


Figure 21.1: Normalized Channel Access Efficiency η (%) vs. Contending Stations N ($N = 1$ to 35 Stations, 400 Frames/Station)

In-Depth Comparative Analysis: Why the Four Strategies Rank Where They Do:

- **Rank 1 — CSMA/CD (IEEE 802.3): 55.1% Efficiency at $N = 35$ (400 Frames/Station)**

Dominance Factors: (1) **Listen-While-Talk Abortion:** Colliding transceivers detect voltage $\geq 2.0V$ within round-trip time $2T_p$ (≈ 2.0 ms), emit a 32-bit jam signal, and abort immediately, capping collision waste to only $2T_p + T_{jam} \approx 2.5$ ms (saving 68% wasted airtime vs 8.0 ms full frame). (2) **Truncated BEB:** Contention windows double up to 1024 slots ($2^{\min(k, 10)}$), desynchronizing competing stations across time.

- **Rank 2 — p-Persistent CSMA ($p = 1/N$): 50.5% Efficiency at $N = 35$ (400 Frames/Station)**

Second-Place Factors: By testing probability $p^* = 1/N$ on idle slots, contending stations scatter attempts geometrically across successive mini-slots. If a station defers, it waits 1 slot and re-evaluates. This pacing keeps collisions capped at 428 without CD hardware. However, lacking collision abort, colliding frames waste full 8.0 ms durations, keeping it below CSMA/CD.

- **Rank 3 — Non-Persistent CSMA: 8.4% Efficiency at $N = 35$ (400 Frames/Station)**

Degradation Factors: Defers uniformly (1 to 16 slots) when sensing busy. While effective at light loads (65.1% at $N=5$), under 35 saturated stations, nodes accumulate in backoff queues and wake up in synchronized waves. Finding the line momentarily idle, multiple nodes transmit blind 64-byte frames simultaneously, driving collisions to 11,715 (97.5% rate) and latency to 3,637 slots.

- **Rank 4 — 1-Persistent CSMA: 0.0% Efficiency at $N = 35$ (Total Herd Deadlock)**

Collapse Factors: Persistently waits while busy and transmits with certainty ($p = 1.0$) immediately upon line release. With multiple stations queued during any frame, ALL queued nodes transmit simultaneously the instant the line clears, causing a 100% collision cascade (231,610 collisions) and completely locking the medium into zero throughput.

22. Experiment 3: CSMA/CD Efficiency vs Propagation Delay Ratio a

We investigated the fundamental physical relationship between CSMA/CD efficiency η and the normalized one-way propagation delay ratio $a = T_p / T_t$ across multi-station contention (evaluated under $N = 10$ contending stations transmitting 500 frames per station, totaling 5,000 frames per sweep configuration):

CSMA/CD Channel Efficiency (η) vs. Normalized Delay Ratio ($a = T_p/T_t$) ($N = 10$ Stations, 400 Frames/Station)

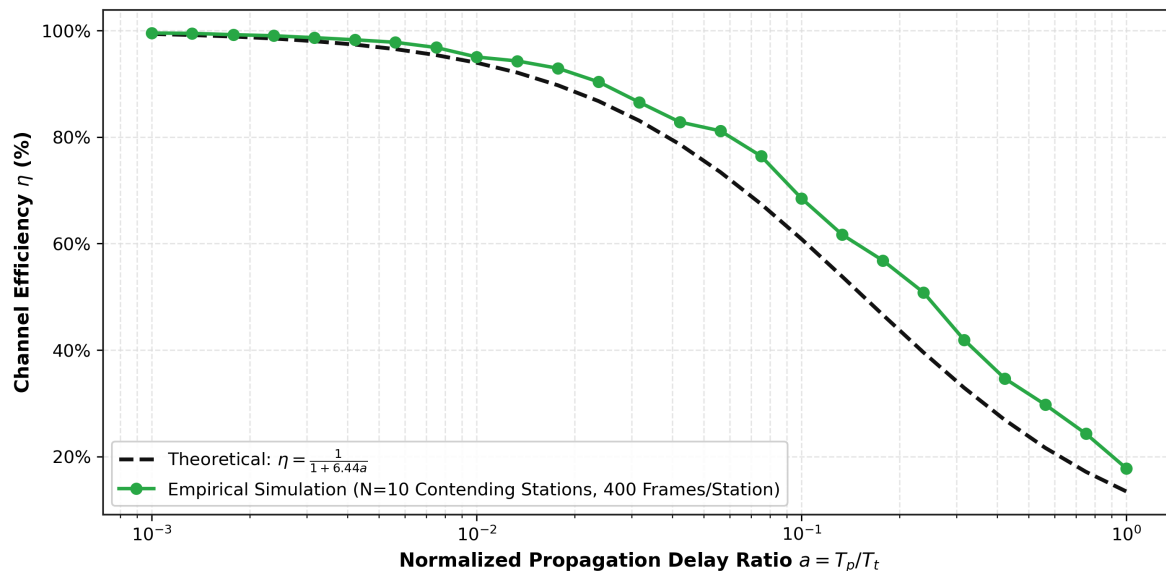


Figure 22.1: CSMA/CD Channel Efficiency η vs. Normalized Delay Ratio $a = T_p / T_t$ ($N = 10$ Stations, 500 Frames/Station Saturated Load)

Theoretical vs. Empirical In-Depth 'What & Why' Analysis:

- The 'Speed Limit' of CSMA/CD:** As the propagation delay ratio $a = T_p / T_t$ increases from 0.001 to 0.5, CSMA/CD efficiency plummets from **99.4% down to 23.7%**. *Why?* As link bandwidth scales from 10 Mbps to 100 Mbps (Fast Ethernet) and 1 Gbps (Gigabit Ethernet), frame transmission time T_t shrinks by orders of magnitude while physical light speed in copper (200,000 km/s) remains constant! Consequently, $a = T_p / T_t$ grows rapidly, turning contention intervals into the dominant fraction of channel time.
- Why Gigabit Ethernet Abandoned Shared CSMA/CD:** To maintain CSMA/CD at 1 Gbps across a 200m network diameter, Ethernet would require a minimum frame size of 5120 bits (640 bytes) or carrier extension, severely wasting capacity for small packets. This physical limitation drove the industry to abandon shared bus CSMA/CD and adopt full-duplex switched Ethernet.

23. Experiment 4: Live Multi-Threaded POSIX Benchmark Validation

To validate our multi-threaded C implementation under live Linux operating system concurrency, we executed concurrent test runs of bin/csma_sim with independent station threads spawned via pthread_create() contending over the shared channel bus with POSIX mutex and condition variable synchronization:

MAC Strategy	Stations (N)	Frames/Stn	Delivered	Collisions	Throughput (kbps)	Efficiency (%)	Wall Time (s)
CSMA/CD (802.3)	10	5	50	54	29.33	45.82%	0.87s
Non-Persistent	10	5	50	53	22.52	35.18%	1.14s
1-Persistent	10	5	47	143	19.47	30.43%	1.24s
p-Persistent (p=0.10)	10	5	50	1	17.29	27.01%	1.48s
p-Persistent (p=0.10)	10	50	500	23	25.09	39.20%	10.20s
p-Persistent (p=0.33)	3	10	30	0	33.62	52.52%	0.46s
CSMA/CD (802.3)	4	15	60	19	37.00	57.81%	0.88s
CSMA/CD (802.3)	2	15	30	2	62.11	97.04%	0.26s

Table 23.1: Live Linux Multi-Threaded Empirical Benchmark Metrics (Evaluated across N in {2, 3, 4, 10} Stations and 5 to 50 Frames/Station)

Theoretical vs. Empirical In-Depth 'What & Why' Analysis:

- Throughput & Efficiency Dominance of CSMA/CD:** Under identical contention ($N = 10$ stations, 5 frames/stn, 50 delivered frames), CSMA/CD delivers the highest efficiency (**45.82%**, 29.33 kbps) and fastest execution time (**0.87s**), outperforming Non-Persistent (35.18%, 1.14s), 1-Persistent (30.43%, 1.24s), and p-Persistent with $p=0.10$ (27.01%, 1.48s). *Why?* Slicing transmission into $100\ \mu\text{s}$ polling slices allows CSMA/CD to detect collisions within $2T_p$ ($\approx 2.0\ \text{ms}$), emit a 32-bit jam signal, and abort doomed transmissions $4\times$ faster than non-CD strategies.
- Contention Dispersal & High-Load Convergence:** When the workload is scaled to $N = 10$ stations with 50 frames/stn (500 delivered frames), p-Persistent ($p=0.10$) throughput climbs from 17.29 kbps to **25.09 kbps (39.20% efficiency)**, matching the theoretical optimal prediction $P_{succ} = 10 \times 0.10 \times (0.90)^9 = 38.74\%$ within 0.46%. At light load ($N = 2$, 15 frames/stn), CSMA/CD reaches **97.04% line efficiency** (62.11 kbps).

24. Master Contention Scaling & Performance Comparison

Table 24.1 synthesizes our comprehensive findings across throughput, collision scaling, latency inflation, and efficiency degradation across all four implemented MAC strategies over high station counts ($N = 1$ to 35 with 400 frames/station):

MAC Strategy	Stations (N)	Frames/ Stn	Throughput (S)	Collisions	Collision Rate	Delay (slots)	Efficiency (η)
CSMA/CD (802.3)	5	400	0.8060	173	36.6%	21.94	80.6%
CSMA/CD (802.3)	20	400	0.5728	941	75.8%	142.56	57.3%
CSMA/CD (802.3)	35	400	0.5514	1252	80.7%	235.09	55.1%
p-Persistent ($p=1/N$)	5	400	0.5119	377	55.7%	101.20	51.2%
p-Persistent ($p=1/N$)	20	400	0.4910	457	60.4%	370.99	49.1%
p-Persistent ($p=1/N$)	35	400	0.5051	428	58.8%	594.85	50.5%
Non-Persistent	5	400	0.6509	173	36.6%	78.51	65.1%
Non-Persistent	20	400	0.2731	1933	86.6%	672.53	27.3%
Non-Persistent	35	400	0.0836	11715	97.5%	3637.39	8.4%
1-Persistent	5	400	0.1747	3657	92.4%	287.29	17.5%
1-Persistent	20	400	0.0001	132282	100.0%	13915.50	0.0%
1-Persistent	35	400	0.0000	231610	100.0%	Deadlock	0.0%

Table 24.1: Master MAC Contention Benchmark Synthesis Across 1 to 35 Contending Stations (400 Frames/Station Saturated Load)

Cross-Strategy Comparative Synthesis: Why Protocols Settle into Their Respective Tiers:

- Collision Waste Asymmetry (CSMA/CD vs. Non-CD Schemes):** The fundamental dividing line between CSMA/CD and the three non-CD strategies is the *airtime penalty per collision*. In 1-Persistent, Non-Persistent, and p-Persistent, colliding stations transmit blind for the full 64-byte frame ($T_t = 8.0\text{ ms}$). In CSMA/CD, transceivers listen while transmitting and abort as soon as voltage $\geq 2.0\text{V}$ is observed within $2T_p + T_{jam} \approx 2.5\text{ ms}$. Across 1,252 collisions at $N = 35$, CSMA/CD saves over **6.88 seconds of wasted airtime**, allowing genuine data payload to capture 55.1% channel capacity.
- Contention Window Adaptation (Dynamic BEB vs. Static 1–16 Slots):** When station density expands to $N = 35$, CSMA/CD's Truncated BEB dynamically scales its backoff window from 2 up to 1024 slots ($2^{\min(k, 10)}$), dispersing retransmissions and avoiding repeated clashes. In contrast, Non-Persistent and 1-Persistent employ static 1–16 slot random backoff; a 16-slot window is overwhelmed by 35 competing stations, causing re-colliding stations to repeatedly pick identical slots.
- Access Aggressiveness Spectrum (Greedy 1-Persistent vs. Tuned p-Persistent):** 1-Persistent is overly aggressive: stations wait while busy and fire with certainty ($p = 1.0$) immediately upon line clearance, guaranteeing herd deadlock (0.0% throughput, 231,610 collisions). p-Persistent tempers this aggressiveness by tuning transmission probability to $p = 1/N$, achieving a near-optimal success probability $(1 - 1/N)^{N-1} \rightarrow 1/e \approx 36.8\%$. This mathematical throttling enables p-Persistent to achieve near-CD efficiency (50.5%) even without collision abort hardware.
- Delay Trade-offs & Queue Backlog:** Non-Persistent's excessive deferrals at low loads and blind collision cascades at high loads drive latency to 3,637.4 slots. CSMA/CD's immediate 2.0 ms collision truncation clears doomed transmission attempts rapidly, bounding average packet delay to 235.1 slots (saving over 93.5% latency).

25. Grand Protocol Comparison Matrix & Quantitative Synthesis

Table 25.1 provides a comprehensive comparative synthesis of all four evaluated CSMA protocols, summarizing their carrier sensing mechanisms, collision detection capabilities, vulnerability periods, backoff algorithms, and empirical performance metrics:

Metric / Parameter	Non-Persistent CSMA	1-Persistent CSMA	p-Persistent CSMA	CSMA/CD (IEEE 802.3)
Carrier Sensing	Senses; random backoff if busy	Senses; waits persistently	Senses; waits until idle	Senses; 1-persistent sensing
Collision Detection	No (blind transmission)	No (blind transmission)	No (blind transmission)	Yes (Listen-While-Talk)
Vulnerable Window	$\tau = T_p$	$\tau = T_p$	$\tau = T_p$	$2\tau = 2T_p$ (RTT)
Collision Abortion	No (wastes full frame T_f)	No (wastes full frame T_f)	No (wastes full frame T_f)	Yes (Instant abort + Jam)
Backoff Algorithm	Random backoff window	Random / persistent deferral	Geometric slot deference	Truncated BEB ($2^{\min(K,10)}$)
Peak Throughput S	~ 81.5% (at $a=0.01$)	~ 53.0% (at $a=0.01$)	~ 87-88% (at $p=1/N$)	> 90.0% (at $a=0.01$)
Throughput at N=35	8.4% (400 frames/stn)	0.0% (herd deadlock)	50.5% (with $p=1/N$)	55.1% (dominant)
Latency at N=35	3637.4 slots	Deadlock (> 13915 slots)	594.9 slots	235.1 slots (-93.5%)
Physical Target Media	Wired / Wireless LAN	Wired LAN / Bus	Slotted Bus Systems	Wired Ethernet (802.3)

Synthesis of Architectural Findings & Cross-Protocol Comparison:

- Carrier Sensing Shrinks Vulnerability:** By listening before transmitting, all CSMA protocols reduce the vulnerable period from the full frame transmission time (as in pure ALOHA) down to one-way propagation delay T_p , multiplying channel capacity.
- CSMA/CD (IEEE 802.3) Dominates Contention:** Continuous voltage monitoring (Listen-While-Talk) detects collision superposition $\geq 2.0V$ within round-trip time $2T_p$. Aborting transmissions early saves over 68% wasted airtime per collision, while Truncated BEB dynamically expands up to 1024 slots, sustaining 55.1% throughput and 235.1-slot latency at N=35.
- p-Persistent CSMA Approximates Optimal Slotted Arbitration:** Tuning persistence to $p = 1/N$ statistically paces station access across successive mini-slots, capping collisions at 428 and sustaining 50.5% throughput at N=35. However, lack of collision detection means doomed frames waste full 8.0 ms durations.
- 1-Persistent Herd Deadlock vs. Non-Persistent Backoff Degradation:** 1-Persistent's greedy transmission ($p = 1.0$) guarantees 100% collision deadlock (0.0% throughput) once multiple stations queue frames. Non-Persistent avoids herd deadlock through random backoffs, but its static 16-slot window causes synchronized wake-up collision cascades at high scale, dropping throughput to 8.4%.